

# Block diagram

sheet symbols open the pages; arrows are the interfaces between them

## Power and ESP32-S3 DevKit

ESP32-S3 DevKitC-1 (N16R8): supply pins, reserved GPIO  
5V0 -> FB1 -> LP5907 -> 3V3\_RADIO -> VMCU\_IN  
DevKit 3V3 decoupling, test points

Fájl: power.kicad\_sch

The ESP32 module is split in four units, each on the page it serves:  
IC1A supply (p2), IC1B radio link (p3), IC1C user interface (p4), IC1D reserved (p2).

## Radio board (BRD4265B)

P200/P201 to the BRD4265B (EFR32FG23)  
FG23 SPI (IO\_MUX slave) + UART + RDY  
SWD bit-bang, I2C board-ID, cut points JP1-JP8

Fájl: radio.kicad\_sch

## User interface

ER-TFTM024-3 TFT, 4-wire SPI + XPT2046 touch  
SSD1306 OLED on I2C 0x3C  
two ADC1 button ladders (6 keys)

Fájl: ui.kicad\_sch

# Mechanical

M3 mounting holes, not plated to GND

H1  
M3  
H2  
M3  
H3  
M3  
H4  
M3

# Design notes

Net names are global (labels and rail symbols); layout, twin and firmware share one name per net.

A label sits on a short stub at a pin end; wires stay inside a block; supplies only as rail symbols.

GPIO plan: FG23 link on the SPI2 IO\_MUX pins 10/11/12 + UART TX 13 + RDY 14; TFT on SPI3 via the  
matrix (4-7, 15-18); touch 8/9/21; buttons ADC1 1/2; I2C 41/42; SWD 38/39/48, RESETn 40.

Forbidden: strapping 0/3/45/46, octal PSRAM 35-37 (N16R8). Reserved: UART0 43/44, USB 19/20, WS2812 47.

Cut points JP1-JP8 isolate every radio-board signal; R1-R5 are OR links on a damping-resistor footprint.

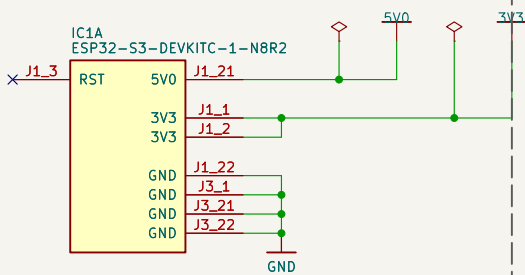
Assembly: R26/R27 DNP, value NA(4.7k). Values carry the critical parameter (value/voltage/tolerance).

FB1: value 600R@100MHz but MPN field BLM18SP300SH1D (30R) - resolve before ordering.

ESP32-S3 DevKitC-1

supply pins: the DevKit is the power source

RST = EN, button on the DevKit

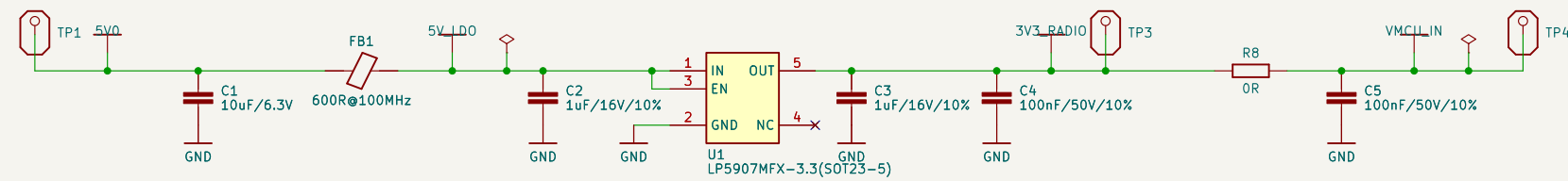


GND test point and ERC flag



Radio supply

5V0 -> FB1 -> LP5907 -> 3V3\_RADIO -> R8 -> VMCU\_IN



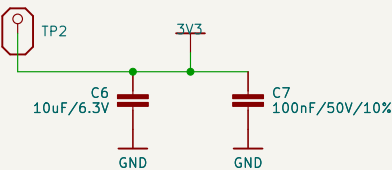
FB1 + C1/C2 keep DevKit 5 V noise out of the LDO

LP5907: 250 mA, 6.5 uVrms, EN tied to IN

R8 OR fitted: VMCU\_IN = 3V3\_RADIO (P201 pin 2)

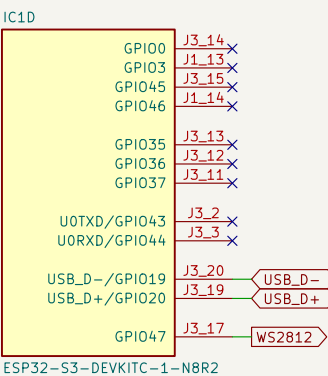
3V3 decoupling

DevKit LDO output: displays and pull-ups



Reserved GPIO

not used on this board - keep it that way



strapping pins (GPIO0/3/45/46): leave open

octal PSRAM of the N16R8 module: forbidden

UART0 = DevKit USB-UART console

native USB on the DevKit connector

RGB LED on the DevKit (AliExpress clone: GPIO47)

DevKit 5V0/3V3 in, 5 V filter, LP5907 radio LDO, reserved GPIO

RF Sparkling Kft. - HA7DCD

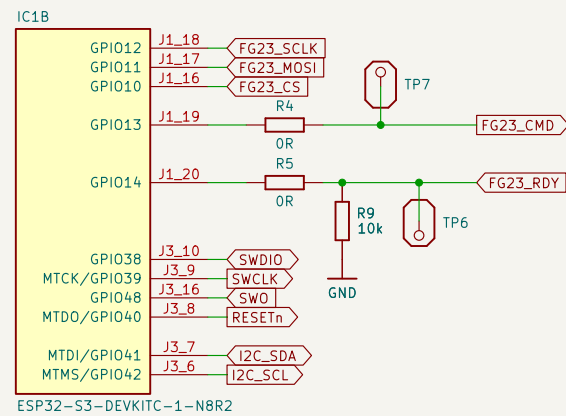
Sheet: /Power and ESP32-S3 DevKit/  
File: power.kicad\_sch

Title: ionos-sdr-mini rev-B3 - Power and ESP32-S3 DevKit

Size: A3 Date: 2026-10-09 Rev: B3  
KiCad E.D.A. 10.0.6 Id: 2/4

ESP32–S3 radio link

SPI2 IO\_MUX slave, UART, SWD, I2C



ESP32 is the SPI slave (FG23 = master), IO\_MUX pins only – no GPIO matrix.

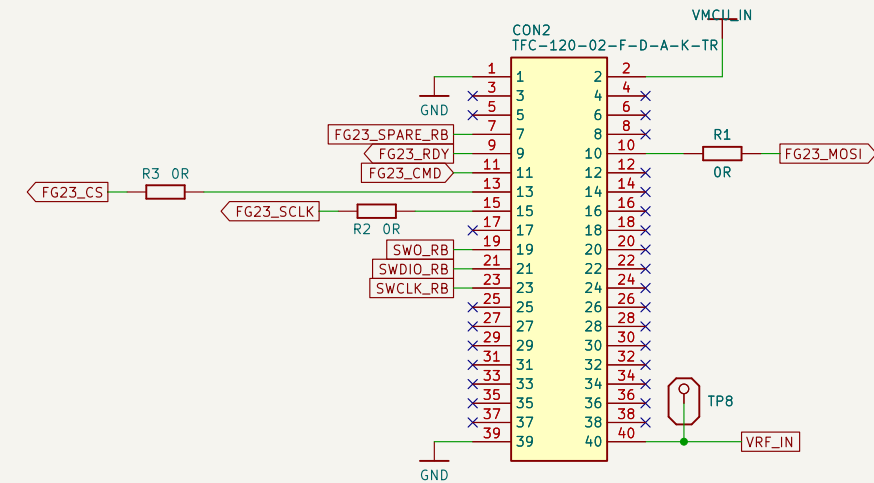
R4/R5 near the ESP32; R9 near P201.

SWD is bit-banged; RESETn open–drain.

Laid out for the AliExpress DevKit clone (LED on GPIO47). Official DevKitC–1 puts the WS2812 on GPIO38 (v1.1) / GPIO48 (v1.0) = SWDIO / SWO here: remove that LED or use the clone.

P201 – radio board header

FG23 link, SWD, VMCU (BRD4265B expansion)



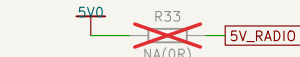
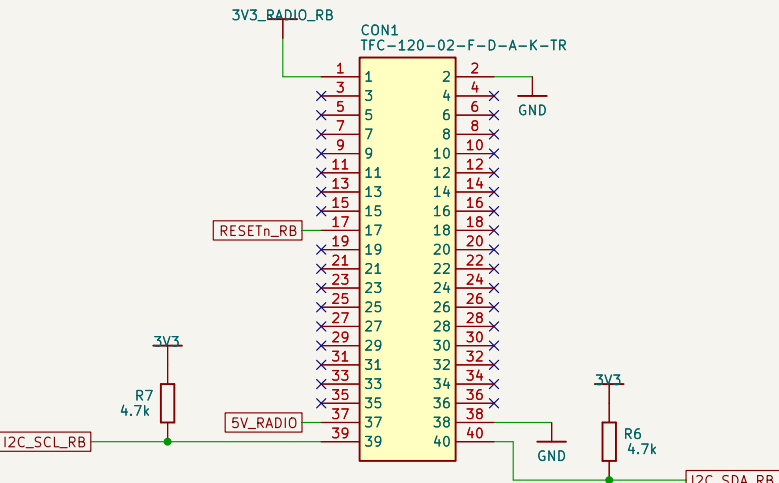
R1–R3 sit at P201: 0R fitted, 22–68R damping footprint (2–layer build: 68R).

Pinout = Silabs BRD4265B / WSTK expansion header.

VRF\_IN (P201.40): track to TP8 only – a test point on the BRD4265B, not connected there.

P200 – radio board header

supply, reset, I2C (board-ID EEPROM)

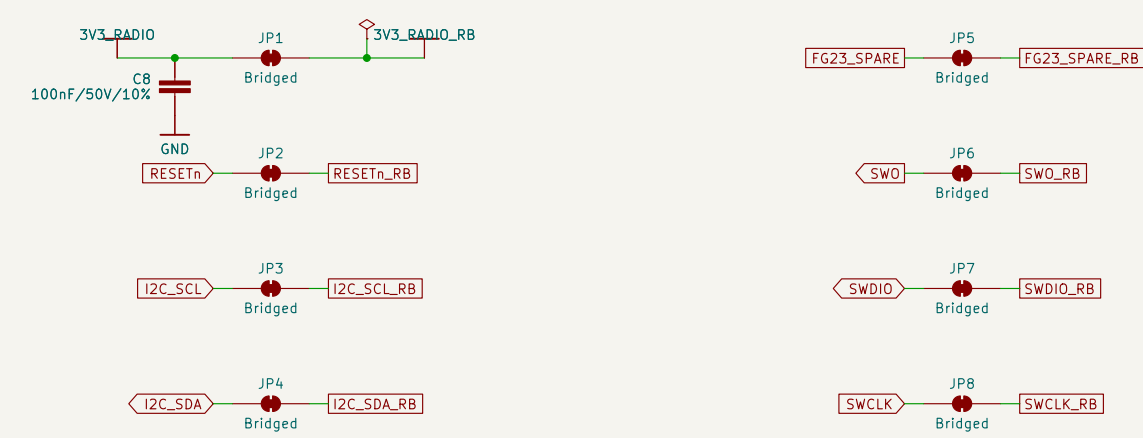


R33 not fitted: DevKit 5 V to P200.37, unused on the BRD4265B.

R6/R7: I2C pull-ups on the radio side. Board-ID EEPROM on the radio board, OLED 0x3C on ours.

Cut points

one solder jumper per radio–board signal, bridged by default



FG23\_SPARE: track and test pad only, no part fitted.

P200/P201 mezzanine, FG23 link, SWD, cut points

RF Sparkling Kft. – HA7DCD

Sheet: /Radio board (BRD4265B)/

File: radio.kicad\_sch

Title: ionos-sdr-mini rev-B3 – Radio board (BRD4265B)

Size: A3

Date: 2026–10–09

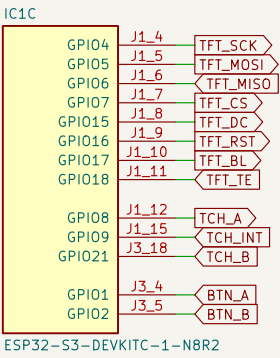
Rev: B3

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Id: 3/4

ESP32–S3 user interface

TFT on SPI3 via the GPIO matrix

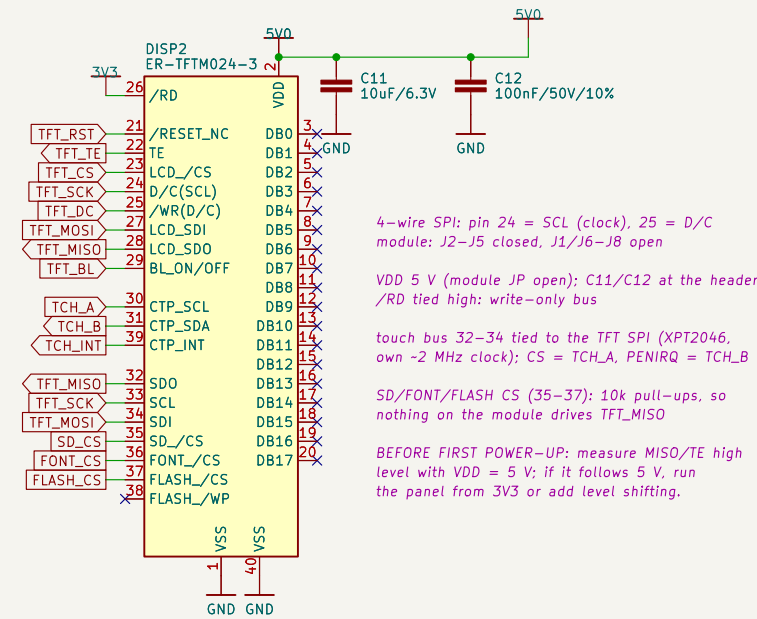


Touch, no jumper – firmware picks the mode:  
TCH\_A GPIO8 resistive CS / CTP SCL  
TCH\_B GPIO21 PENIRQ / CTP SDA  
TCH\_INT GPIO9 unused / CTP INT  
R31/R32 4.7k fitted in every build.

BTN\_A/BTN\_B: ADC1, usable with WiFi on.

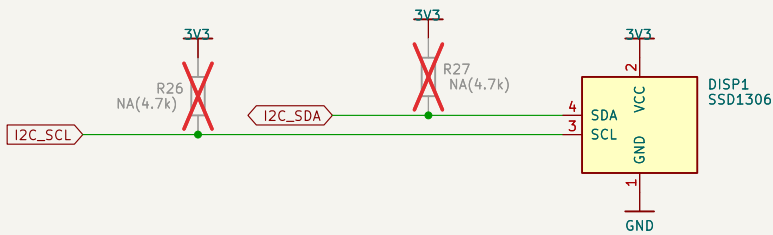
TFT ER–TFTM024–3

2.4 in 240x320, 4–wire SPI, resistive touch (XPT2046)



OLED SSD1306

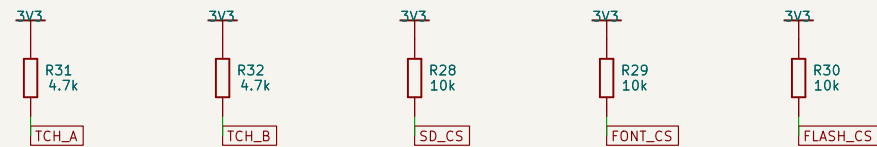
0.96 in 128x64, I2C 0x3C



R26/R27 not fitted: the bus pull–ups are R6/R7 on the radio side; fit them if JP3/JP4 are cut.

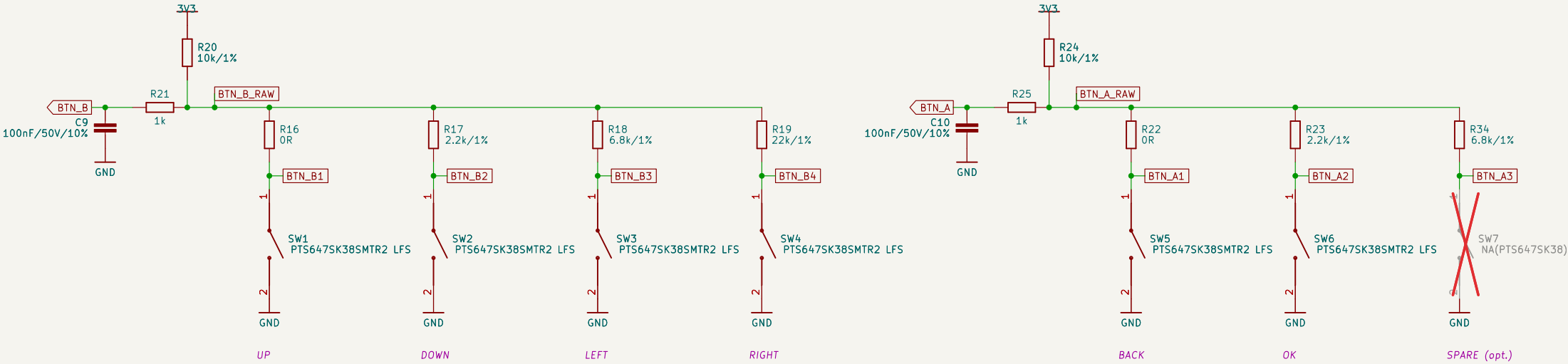
Header pull–ups

touch CS/IRQ and the module flash/font/SD chip selects



Buttons

two resistor ladders on ADC1: one GPIO per ladder



Both ladders: OR / 2.2k / 6.8k / 22k under a 10k pull–up → one firmware table:  
levels 0 / 0.60 / 1.34 / 2.27 / 3.3 V, thresholds 0.30 / 0.97 / 1.80 / 2.79 V. Wake: OR keys only.

TFT + resistive touch, OLED, two ADC button ladders

RF Sparkling Kft. – HA7DCD

Sheet: /User interface/

File: ui.kicad\_sch

Title: ionos–sdr–mini rev–B3 – User interface

Size: A3

Date: 2026–10–09

Rev: B3

KiCad E.D.A. 10.0.6

Id: 4/4